

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL NO 2
SUBJECT - AI

AIM - Recursive Best-First Search Algorithm (RBFS)

CODE -

```
import matplotlib.pyplot as plt
```

```
graph = {  
    "Andheri East": {  
        "Bandra": 12,  
        "Sion": 14,  
        "Jogeshwari": 10  
    },  
  
    "Bandra": {  
        "Worli": 10  
    },  
  
    "Worli": {  
        "Haji Ali": 5  
    },  
  
    "Haji Ali": {  
        "Babulnath Mandir": 4  
    },  
  
    "Sion": {  
        "Byculla": 10  
    },  
  
    "Byculla": {  
        "Marine Lines": 10  
    },  
  
    "Marine Lines": {  
        "Babulnath Mandir": 6  
    },  
  
    "Jogeshwari": {  
        "Dadar": 12  
    },  
  
    "Dadar": {  
        "Grant Road": 10  
    },  
  
    "Grant Road": {  
        "Babulnath Mandir": 8  
    },  
  
    "Babulnath Mandir": {}  
}
```

```
heuristics = {  
    "Andheri East": 41,
```

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ROLL NO - T109

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL NO 2
SUBJECT - AI

```
"Bandra": 25,  
"Worli": 12,  
"Haji Ali": 4,  
"Sion": 28,  
"Byculla": 16,  
"Marine Lines": 6,  
"Jogeshwari": 30,  
"Dadar": 18,  
"Grant Road": 8,  
"Babulnath Mandir": 0  
}  
  
coords = {  
    "Andheri East": (1,5),  
    "Bandra": (3,7),  
    "Worli": (5,7),  
    "Haji Ali": (7,7),  
    "Sion": (3,5),  
    "Byculla": (5,5),  
    "Marine Lines": (7,5),  
    "Jogeshwari": (3,3),  
    "Dadar": (5,3),  
    "Grant Road": (7,3),  
    "Babulnath Mandir": (9,5)  
}  
  
def rbfs_search(start, goal):  
    success, path, cost, _ = rbfs(  
        start,  
        goal,  
        g=0,  
        f_limit=float('inf'),  
        path=[start]  
    )  
    return path, cost  
  
def rbfs(node, goal, g, f_limit, path):  
  
    if node == goal:  
        return True, path, g, g  
  
    neighbours = graph[node]  
  
    if not neighbours:  
        return False, [], 0, float('inf')  
  
    successors = []  
  
    for neighbour, distance in neighbours.items():  
  
        if neighbour not in path:
```

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ROLL NO - T109

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PRACTICAL NO 2
SUBJECT - AI

```
    next_g = g + distance
    next_f = max(next_g + heuristics[neighbour], g + heuristics[node])

    successors.append([next_f, neighbour, next_g])

if not successors:
    return False, [], 0, float('inf')

while True:

    successors.sort(key=lambda x: x[0])

    best = successors[0]

    if best[0] > f_limit:
        return False, [], 0, best[0]

    alternative = successors[1][0] if len(successors) > 1 else float('inf')

    success, result_path, total_g, returned_f = rbfs(
        best[1],
        goal,
        best[2],
        min(f_limit, alternative),
        path + [best[1]]
    )

    best[0] = returned_f

    if success:
        return True, result_path, total_g, returned_f

path, total_dist = rbfs_search(
    "Andheri East",
    "Babulnath Mandir"
)

print("Optimal RBFS Path:")
print(" -> ".join(path))
print("Total Distance:", total_dist, "km")

plt.figure(figsize=(12,7))

for node, neighbours in graph.items():

    x1, y1 = coords[node]

    for neighbour, dist in neighbours.items():

        x2, y2 = coords[neighbour]

        is_path = (
```

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ROLL NO - T109

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL NO 2
SUBJECT - AI

```
        node in path and
        neighbour in path and
        path.index(neighbour) == path.index(node)+1
    )

    color = "green" if is_path else "gray"
    width = 3 if is_path else 1.5

    plt.plot(
        [x1, x2],
        [y1, y2],
        color=color,
        linewidth=width
    )

    plt.text(
        (x1+x2)/2,
        (y1+y2)/2,
        str(dist)+" km",
        color="red",
        fontsize=8
    )

for node, (x, y) in coords.items():

    color = "gold" if node in path else "skyblue"

    plt.scatter(
        x,
        y,
        s=700,
        color=color,
        edgecolors="black"
    )

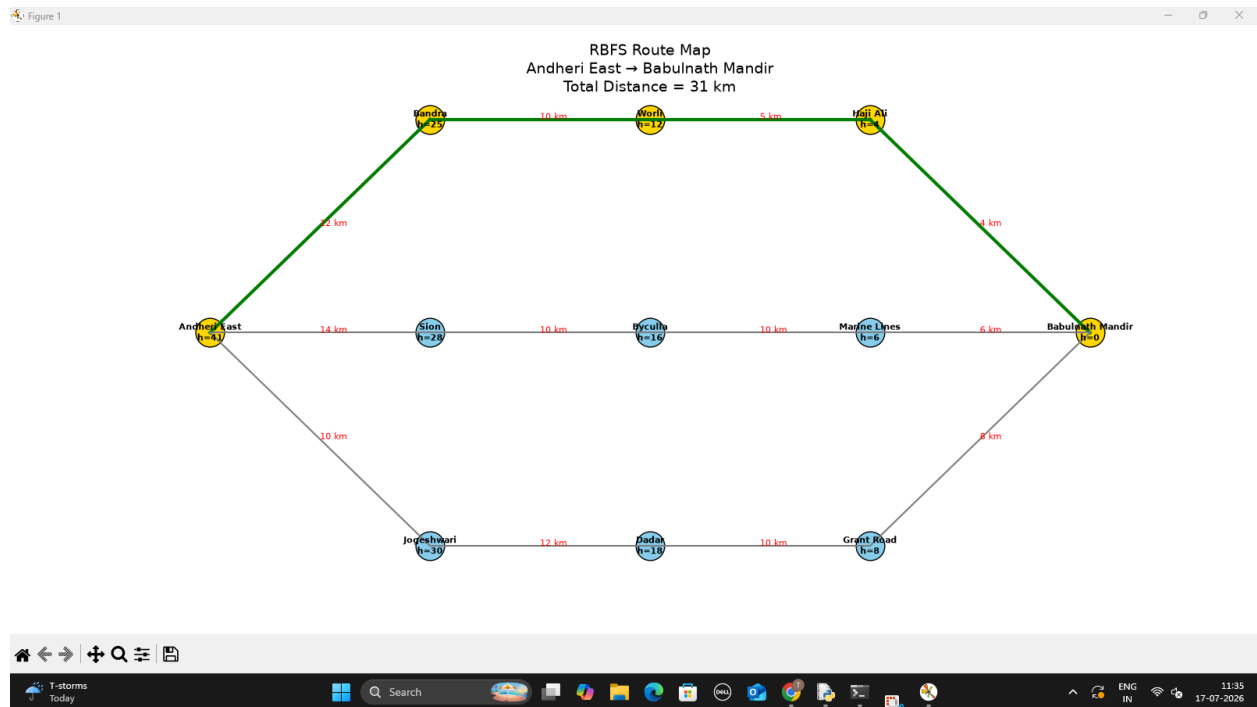
    plt.text(
        x,
        y,
        f"{node}\nh={heuristics[node]}",
        ha="center",
        va="center",
        fontsize=8,
        color="black",
        fontweight="bold"
    )

plt.title(f"RBFS Route Map\nAndheri East → Babulnath Mandir\nTotal Distance = {total_dist} km", fontsize=14)

plt.axis("off")
plt.show()
```

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PRACTICAL NO 2
SUBJECT - AI



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